

Ares Koumbliis

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Pittsburgh, PA, USA

Aug 2028

National Technical University of Athens

Integrated Master's (BE/ME) in Electrical and Computer Engineering

Athens, Greece

Feb 2024

GPA: 9.3/10.0

SKILLS

Programming Languages: Python, C++, C

Software: PyTorch, OpenCV, Scikit-Learn, Git, Linux, ROS

Languages: English, Greek

PROFESSIONAL EXPERIENCE

Tesla

Palo Alto, CA, USA

Senior Autopilot Software Development Engineer

Aug 2025 – Jul 2026

Autopilot Software Development Engineer

Feb 2025 – Aug 2025

Autopilot Software QA Engineer

Jan 2024 – Feb 2025

- Fine-tuned end-to-end driving policies using reinforcement learning, improving safety in low-visibility conditions without regressing ride comfort
- Enhanced explicit trajectory selection logic, enabling the release of FSD v14 on HW3 vehicles and in left-hand-traffic markets
- Implemented reduced-visibility slowdown logic across vehicle platforms, reducing collision rate in simulation by 44%
- Adapted Lane Departure Avoidance for rear-wheel-steer vehicle dynamics and deployed it on Cybertruck as prerequisite for 5-star NCAP
- Led validation of Tesla's closed-loop video-generation simulator, demonstrating strong correlation with both track-test results and real-world miles per critical intervention
- Developed safety and comfort evaluation metrics for Tesla's open-loop, closed-loop, and video-generation simulators, enabling early rejection of underperforming models and reducing reliance on on-vehicle testing
- Led platform-specific validation and release efforts for first FSD deployment on Cybertruck (~30,000 vehicles)

Alexander Moore SA

Athens, Greece

Data Science Intern

Mar 2021 – Jul 2021

- Built a regression model to forecast product demand, reducing prediction error by 50% over existing approaches
- Developed a neural network using tf-idf features to classify customer feedback as positive or negative, achieving 96% accuracy
- Designed a recommendation system based on customer similarity for a pharmaceutical supplier

ACADEMIC PROJECTS

PROM Racing NTUA (FSAE)

Athens, Greece

National Technical University of Athens

Oct 2021 – Aug 2023

- Founded and led a 10-member team to develop and deliver an autonomous race car over a 2-year timeline
- Fine-tuned and deployed a YOLOv5 object detection model on TPU for real-time track cone detection
- Designed, trained, and deployed a ResNet-based keypoint detector on iGPU for accurate cone distance estimation
- Developed a YOLOv8 variant with a regression head for direct cone distance estimation using knowledge distillation from the existing perception stack
- Implemented an Extended Kalman Filter fusing IMU, wheel-speed, and steering data for robust vehicle state estimation
- Built a graph-based SLAM system using the iSAM algorithm for real-time cone and vehicle localization
- Developed a Koopman MPC framework for path tracking, leveraging an encoder-decoder network to capture nonlinear vehicle dynamics